



11672 - Greedy Big Brother: examining the atmosphere of a Neptune-sized outer companion to an unusual sub-Neptune

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NIRISS	NIRISS Single-Object Slitless Spectroscopy	(1) TOI-421
	2	NIRSPEC	NIRSpec Bright Object Time Series	(1) TOI-421

ABSTRACT

We propose to acquire a 0.9-5.1m transmission spectrum of TOI-421 c, a 600 K Neptune-sized exoplanet orbiting a G9 star, by observing two transits: one with NIRISS/SOSS and one with NIRSpec/G395H. A 900 K sub-Neptune in the same system, TOI-421 b, was observed with JWST and was found to host a cloud-free, low-metallicity ($<10\times$ solar) atmosphere. This starkly contrasts with other sub-Neptunes observed using JWST, with

JWST Proposal 11672 (Created: Friday, May 29, 2026, 7:00:12PM Eastern Standard Time) - Overview

typical atmospheric metallicities $>100\times$ solar. Our observations will determine whether TOI-421 c, the larger, outer planet in the system, is the cause of TOI-421 b's unusual atmosphere. By measuring the atmospheric metallicity of TOI-421 c, we will test the theory of pebble filtering, which could explain the low metallicity of TOI-421 b. We will also use our atmospheric measurements to constrain the core mass of TOI-421 c, which we will directly compare to that of TOI-421 b. Furthermore, this system provides a unique opportunity to ascertain whether cloud and haze formation preferentially occurs on sub-Neptune to Neptune-sized planets at intermediate temperatures (500-800 K).

Our proposal will have a high impact both in the field of exoplanet atmospheres, by enabling comparative planetology, and in planet formation, by helping to understand the origin of this multi-planet system. The TOI-421 system provides a rare opportunity to probe a representative case from the numerous FGK star multi-planet systems discovered by facilities such as Kepler, and enables comparative studies with M dwarf sub-Neptune systems, of which many already have scheduled or executed JWST observations.

OBSERVING DESCRIPTION

We propose a time series transmission spectroscopy of a transiting exo-Neptune using NIRISS SOSS and NIRSpec G395H. The planet has an orbital period of 16.067541 days and transit duration of 2.755 hours. We choose out-of-transit baseline equal to the transit duration and allow 45 minutes for detector settling.

The NIRSpec target acquisition has to be done on a nearby star because the target is too bright.

We will use SUBSTRIP95 for NIRISS. JWST ETC allows for 3 groups per integration without saturation.

We will use SUB2048 subarray for NIRSPEC. JWST ETC allows for 4 groups per integration without saturation.

Proposal 11672 - Targets - Greedy Big Brother: examining the atmosphere of a Neptune-sized outer companion to an unusual sub-Ne...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	TOI-421	RA: 05 27 24.8259 (81.8534413d) Dec: -14 16 37.05 (-14.27696d) Equinox: J2000	Proper Motion RA: -35.743 mas/yr Proper Motion Dec: 50.387 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Exoplanet Systems, G dwarfs]				
Fixed Targets	(2)	Gaia-DR3- 2984582227215748224	RA: 05 27 23.7421 (81.8489254d) Dec: -14 16 12.21 (-14.27006d) Equinox: J2000	Proper Motion RA: -36.19 mas/yr Proper Motion Dec: 50.607 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[K stars]				

Proposal 11672 - Observation 1 - Greedy Big Brother: examining the atmosphere of a Neptune-sized outer companion to an unusual s...

Observation	Proposal 11672, Observation 1: NIRISS									Sat May 30 00:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRISS Single-Object Slitless Spectroscopy									
Diagnostics	(NIRISS (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(1)	TOI-421	RA: 05 27 24.8259 (81.8534413d) Dec: -14 16 37.05 (-14.27696d) Equinox: J2000		Proper Motion RA: -35.743 mas/yr Proper Motion Dec: 50.387 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Exoplanet Systems, G dwarfs]									
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	SOSSFAINT	F480M	NISRAPID	3	1	1	0.202	304253.2
Template	Subarray					Include Short First Exposure and F277W Exposure?				
	SUBSTRIP96					true				
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	NISRAPID	1	1	1	1	1	4.448		
	2	NISRAPID	3	2944	1	1	2944	26132.357	304253.1	
	3	NISRAPID	3	20	1	1	20	177.53		
Special Requirements	Phase 0.9883175444622588 to 0.9909107643374511 with period 16.067541 Days and zero-phase 2459195.307 HJD Aperture PA Range 30 to 50 Degrees (V3 29.43873283 to 49.43873283) Aperture PA Range 75 to 94 Degrees (V3 74.43873283 to 93.43873283) Aperture PA Range 258 to 276 Degrees (V3 257.43873283 to 275.43873283) Aperture PA Range 292 to 323 Degrees (V3 291.43873283 to 322.43873283) Time Series Observation No Parallel Attachments									

Proposal 11672 - Observation 2 - Greedy Big Brother: examining the atmosphere of a Neptune-sized outer companion to an unusual s...

Observation	Proposal 11672, Observation 2: NIRSPEC										Sat May 30 00:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
	Comments: Target acquisiton is done on a nearby star because the target is too bright. We achieve an S/N of 55.										
Diagnostics	(NIRSPEC (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#										
	Name										
	Target Coordinates										
	Targ. Coord. Corrections										
Fixed Targets	Miscellaneous										
	(1)										
	TOI-421										
	RA: 05 27 24.8259 (81.8534413d)										
Dec: -14 16 37.05 (-14.27696d)											
Equinox: J2000											
Epoch of Position: 2000											
Fixed Targets	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Star										
	Description=[Exoplanet Systems, G dwarfs]										
Acquisition	#										
	Target										
	TA Method										
Acquisition	Subarray										
	Filter										
	Readout Pattern										
Acquisition	Groups/Int										
	Integrations/Exp										
	Total Integrations										
Acquisition	Total Exposure Time										
	Optional ETC ID										
Template	Subarray										
	SUB2048										
Spectral Elements	#										
	Grating/Filter										
	Readout Pattern										
Spectral Elements	Groups/Int										
	Integrations/Exp										
	Exposures/Dith										
Spectral Elements	Total Dithers										
	Total Integrations										
	Total Exposure Time										
Spectral Elements	Optional ETC ID										
Special Requirements	Phase 0.9883175444622588 to 0.9909107643374511 with period 16.067541 Days and zero-phase 2459195.307 HJD										
	Time Series Observation										
Special Requirements	No Parallel Attachments										